

Python – Matplotlib: Plot Your Data

A Beginner's Lesson in Python Data Visualisation

Subject	Level	Duration
Python / Matplotlib	Beginner (Class 4–6)	~60 Minutes

Learning Objectives

By the end of this lesson, students will be able to:

- Define **data** and give real-life examples from everyday Indian life.
 - Explain why we **visualise data** using charts and graphs.
 - Label and read the parts of a **bar chart** (title, X-axis, Y-axis, bars).
 - Describe what **Python** is and why it is used.
 - Explain what **Matplotlib** is in simple terms.
 - Write a **7-line Python program** to draw a bar chart using **plt.bar()**.
 - Identify and fix common **coding errors** (missing commas, quotes, typos).
 - Predict what happens to a chart when values or labels are changed.
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Section 1 — What is Data?

Data is a collection of facts, numbers, or names gathered from the world around us. Every time you count something, measure something, or record something, you are creating data.

Classroom Example

Shreya asked her five friends how many glasses of water they drink every day. She collected the following data:

Name	Glasses of Water / Day
Riya	6
Arjun	4
Meena	8
Karthik	5
Divya	7

Data Is All Around Us

- Test scores in exams
- Daily temperature readings

- Cricket runs scored by players in a match
- School attendance records
- Rainfall amounts across different Indian cities

■ **Key Idea:** Data by itself is just a list of facts. The magic happens when we turn data into a picture — that is called **data visualisation!**

Section 2 — Why Visualise Data?

Our brain processes images **60,000 times faster** than plain text. A well-made chart can communicate information in seconds that would take many sentences to explain.

Rainfall Comparison Example

Compare how much easier it is to understand this data as a chart instead of a list:

City	Rainfall (mm)	Relative Amount
Mumbai	2,207	■■■■■■■■■■■■■■■■■■■■ Highest
Chennai	1,400	■■■■■■■■■■ High
Bengaluru	970	■■■■■■ Medium
Delhi	797	■■■■ Lowest

■■ **Look at the bar for Mumbai** — it is clearly the tallest. You can see this instantly in a chart, but in a text list you have to read and compare every number.

Section 3 — Anatomy of a Bar Chart

A bar chart has five key parts. Knowing each part helps you read *and* build charts correctly.

Title	Tells the reader what the chart is about. Always include one!
X-Axis	The horizontal line at the bottom. It shows the names or categories . Memory tip: " <i>X is the floor.</i> "
Y-Axis	The vertical line on the side. It shows the numbers or values . Memory tip: " <i>Y goes upYard.</i> "
Bars	Rectangles that grow upward from the X-axis. A taller bar = a bigger number . Shorter bar = smaller number.
Values	The numbers shown on or beside each bar so you can read exact amounts without guessing.

Quick-Check Questions (Reading a Bar Chart)

Question	Answer
Which sport got the most votes?	Cricket — it has the tallest bar.
How many students chose Badminton?	11 students.
Which got fewer votes: Kabaddi or Football?	Football (7 votes vs Kabaddi's 9 votes).

Section 4 — What is Python?

Python is a programming language — a special language for giving instructions to a computer. Just like we use English to talk to people, we use Python to talk to computers.

Why Python?

Easy to Read	Python looks almost like plain English. Even beginners can understand it quickly.
Widely Used	Big organisations in India use it: ISRO, IITs, Infosys, TCS, and Flipkart.
Powerful	With just a few lines you can draw charts, build websites, or run data analysis.
Free	Python is completely free to download and use for everyone.

Section 5 — What is Matplotlib?

Matplotlib is a special Python *library* (a toolkit of ready-made tools) for drawing charts and graphs. Think of it as a magical geometry box that helps you make colourful pictures using code instead of pencils and rulers.

■ When we write **`import matplotlib.pyplot as plt`**, we are opening the Matplotlib toolbox and giving it a short nickname — **`plt`** — so we can type less later.

Section 6 — Python Lists

Before we can draw a chart, we need to store our data in **Python lists**. A list is an ordered collection of items enclosed in square brackets **[]**.

Square brackets	Always start and end a list with [] .
Commas	Separate each item with a comma .
Text in quotes	Text (strings) must go inside "double quotes" : e.g. "Cricket" .
Numbers – no quotes	Numbers are written as-is with no quotes: e.g. 14 .

Order matters

The first name always pairs with the first number, second with second, and so on.

Example Lists:

```
sports = ["Cricket", "Kabaddi", "Badminton", "Football", "Kho-Kho"]  
votes  = [14,          9,          11,          7,          5          ]
```

Notice how "Cricket" (index 0) pairs with 14 votes (index 0), "Kabaddi" pairs with 9, and so on.

Section 7 — The 7-Line Code

Here is the complete program that creates a bar chart of favourite sports in Class 4. It is only 7 meaningful lines of code!

```
import matplotlib.pyplot as plt

sports = ["Cricket", "Kabaddi", "Badminton", "Football", "Kho-Kho"]
votes  = [14, 9, 11, 7, 5]

plt.bar(sports, votes)
plt.title("Favourite Sports in Our Class")
plt.show()
```

Line-by-Line Explanation

Line	Code	What it does
1	<code>import matplotlib.pyplot as plt</code>	Opens the Matplotlib chart toolbox and gives it the short name plt .
2	<code>(blank line)</code>	Blank lines make code easier to read — they are not commands.
3	<code>sports = [...]</code>	Creates a list of sport names that will appear on the X-axis .
4	<code>votes = [...]</code>	Creates a list of vote counts that will appear on the Y-axis .
5	<code>(blank line)</code>	Another blank line for readability.
6	<code>plt.bar(sports, votes)</code>	Draws one coloured bar for each sport-vote pair.
7	<code>plt.title("...")</code>	Adds the title at the top of the chart.
8	<code>plt.show()</code>	Displays the finished chart on screen.

Section 8 — Common Coding Errors

Every programmer makes mistakes — even professionals! Learning to spot and fix errors (called **debugging**) is an important skill.

■ Missing comma between list items

Wrong:

```
sports = ["Cricket" "Kabaddi"]
```

Correct:

```
sports = ["Cricket", "Kabaddi"]
```

Python cannot tell where one item ends and the next begins.

■ Forgetting quotation marks around text

Wrong:

```
sports = [Cricket, Kabaddi]
```

Correct:

```
sports = ["Cricket", "Kabaddi"]
```

Without quotes Python thinks Cricket is a variable, not a word.

■ Spelling mistake in a function name

Wrong:

```
plt.Bar(sports, votes) # capital B
```

Correct:

```
plt.bar(sports, votes)
```

Python is case-sensitive. plt.Bar and plt.bar are different things.

■ Unequal list lengths

Wrong:

```
sports = ["Cricket", "Kabaddi"] # 2 items  
votes  = [14, 9, 11]           # 3 items
```

Correct:

```
sports = ["Cricket", "Kabaddi"]  
votes  = [14, 9]
```

Both lists must have the same number of items.

Section 9 — Challenges & Predictions

Great programmers do not just write code — they **predict** what will happen before they run it. Try these two challenges:

Challenge 1 — Change a Value

Change the vote count for Kho-Kho from **5** to **15**.

```
# Original
votes = [14, 9, 11, 7, 5]

# After change
votes = [14, 9, 11, 7, 15]  # Kho-Kho now has 15 votes
```

Your Prediction	Result
What do you think happens to the Kho-Kho bar?	Kho-Kho becomes the tallest bar on the chart! The bar height is controlled entirely by the number in the votes list.

Challenge 2 — Change a Label

Change the label "**Football**" to "**Hockey**".

```
# Original
sports = ["Cricket", "Kabaddi", "Badminton", "Football", "Kho-Kho"]

# After change
sports = ["Cricket", "Kabaddi", "Badminton", "Hockey", "Kho-Kho"]
```

Your Prediction	Result
What changes and what stays the same?	Only the label on the X-axis changes to Hockey. The bar height stays the same at 7 votes — because the votes list is unchanged.

Section 10 — Real-World Uses of Data Visualisation

Data visualisation is not just for school projects — it is used by professionals across many fields every single day.

■ ISRO	Scientists track rocket trajectories, satellite orbits, and mission data on charts to make split-second decisions.
■ Doctors & Hospitals	Medical teams chart a patient's heart rate, blood pressure, and temperature over time to spot problems early.
■ Sports Analysts	Cricket coaches study batting and bowling statistics visually to plan strategies for upcoming matches.
■ Weather Forecasters	Meteorologists display rainfall, temperature, and wind data as charts so the public can understand forecasts easily.
■ Businesses	Companies chart sales, costs, and customer data to decide where to invest and where to cut costs.

Section 11 — Bonus Practice Data Sets

Use these data sets to practise writing more bar chart programs on your own!

Bonus Dataset 1: Favourite Indian Breakfast Foods

Food	Votes	Winner?
Dosa	10	
Idli	8	
Roti	12	■ Winner!
Poha	6	
Upma	7	

```

foods = ["Dosa", "Idli", "Roti", "Poha", "Upma"]
votes = [10, 8, 12, 6, 7]
plt.bar(foods, votes)
plt.title("Favourite Breakfast Foods")
plt.show()

```

Bonus Dataset 2: Favourite School Subjects

Subject	Votes	Winner?
Maths	15	■ Winner!
Science	12	
English	9	
Hindi	11	
Social Studies	8	

```

subjects = ["Maths", "Science", "English", "Hindi", "Social Studies"]
votes    = [15, 12, 9, 11, 8]
plt.bar(subjects, votes)
plt.title("Favourite School Subjects")
plt.show()

```

Bonus Dataset 3: Annual Rainfall in Indian Cities (mm)

City	Rainfall (mm)
Mumbai	2,207
Chennai	1,400
Bengaluru	970
Delhi	797

```

cities    = ["Mumbai", "Chennai", "Bengaluru", "Delhi"]
rainfall  = [2207, 1400, 970, 797]
plt.bar(cities, rainfall)
plt.title("Annual Rainfall in Indian Cities (mm)")
plt.show()

```

Section 12 — Homework Assignment

■ **Your Task:** Think of something fun to count in your home or neighbourhood. Collect **5 names** and **5 numbers**, then write the Python code to draw your own bar chart. Bring your data and code to the next class!

Ideas to Get You Started

- Favourite foods in your family (ask each person)
- Number of books each friend has read this month
- Types of vehicles that pass your house in 10 minutes
- How many hours each family member sleeps
- Favourite colours in your class

Planning Sheet

Names (X-axis)	Numbers (Y-axis)
1.	
2.	
3.	
4.	
5.	

Chart Title I will use: _____

Section 13 — Key Concepts Recap

Use this table as a quick-reference guide when you are writing code.

Term	Definition	Example
Data	A collection of facts, numbers, or names.	Votes for sports in Class 4
Data Visualisation	Turning data into a picture or chart.	A bar chart of favourite sports
Bar Chart	A chart that uses bars to compare values.	Taller bar = more votes
X-axis	Horizontal line — shows names/categories.	Cricket, Kabaddi, Badminton...
Y-axis	Vertical line — shows numbers/values.	0, 5, 10, 15...
Python	An easy-to-read programming language.	Used by ISRO, IITs, Infosys
Matplotlib	Python library for drawing charts.	<code>import matplotlib.pyplot as plt</code>
<code>plt.bar()</code>	Function that draws bars in a chart.	<code>plt.bar(sports, votes)</code>
<code>plt.title()</code>	Function that adds a title to the chart.	<code>plt.title("My Chart")</code>
<code>plt.show()</code>	Function that displays the chart.	Always the last line
List	An ordered collection of items in <code>[]</code> .	<code>[14, 9, 11, 7, 5]</code>
Debugging	Finding and fixing errors in code.	Checking for missing commas

What's Next? — Session Preview

In the next session you will level up your Matplotlib skills:

- **Collect your own data:** Survey friends and family to build a completely original dataset.
- **Add colours to bars:** Use the `color=` parameter to make each bar a different colour.
- **Label the axes:** Add axis labels using `plt.xlabel()` and `plt.ylabel()`.
- **Add value labels:** Show the exact number on top of each bar.
- **Explore other chart types:** Try line charts (`plt.plot()`) and pie charts (`plt.pie()`).

You have learned what data is, how to read and create bar charts, and how to write a Python program to draw them using Matplotlib. Keep practising — every expert was once a beginner!